

1. Introduction:

Light significantly influences human emotion and behaviour. Circadian rhythms, sleep patterns, and mood can all be affected by different intensities and colours of light. This project explores the physiological effects of light on emotional states by analysing bioelectrical signals using non-invasive electrophysiology tools. The setup employs the **BioAmp EXG Pill** to acquire EEG/EMG/ECG data, an **ADS1115 analog-to-digital converter** for high-resolution signal sampling, and a **Raspberry Pi 3A+** for data processing and visualization. Our aim is to correlate variations in bio-signals with controlled light stimuli to interpret emotional responses.

We intend to implement our project in:

- To propose optimized lighting conditions for enhanced emotional well-being.
- Adaptive lighting that changes colour temperature and intensity based on user mood.
- Dynamic lighting in offices to minimize stress and increase efficiency.
- Light therapy for treating Seasonal Affective Disorder (SAD) and depression.
- Hospital lighting optimization to improve patient recovery rates and comfort.
- Mood-based lighting in stores to influence customer emotions and buying behaviour.
- Smart fitting room lighting that adapts to enhance clothing perception.

The novelty of your project lies in its integration of real-time emotional analysis with ambient light modulation using low-cost, compact hardware. Unlike conventional studies that rely on subjective self-reporting or expensive medical-grade systems, your setup utilizes non-invasive, accessible electrophysiological monitoring (BioAmp EXG Pill) and a Raspberry Pi-based processing unit to create a portable, scalable emotional-response detection system. Furthermore, your approach to correlating specific light stimuli (intensity and color) with EEG/EMG/ECG patterns provides a novel pathway for studying the neurophysiological basis of light-induced mood changes in real-world environments.

3. Working Principle:

a) Signal Acquisition:

The BioAmp EXG Pill captures electrophysiological signals (e.g., EEG for brain activity) from the subject under varying light conditions.

b) Signal Amplification and Conversion:

The raw analog signals are amplified by the BioAmp and digitized using the 16-bit ADS1115 ADC to ensure precision before sending to the Raspberry Pi.

c) Processing and Analysis:

The Raspberry Pi processes the digital signal to extract relevant features and correlates these with the type and intensity of light stimuli to infer emotional states.

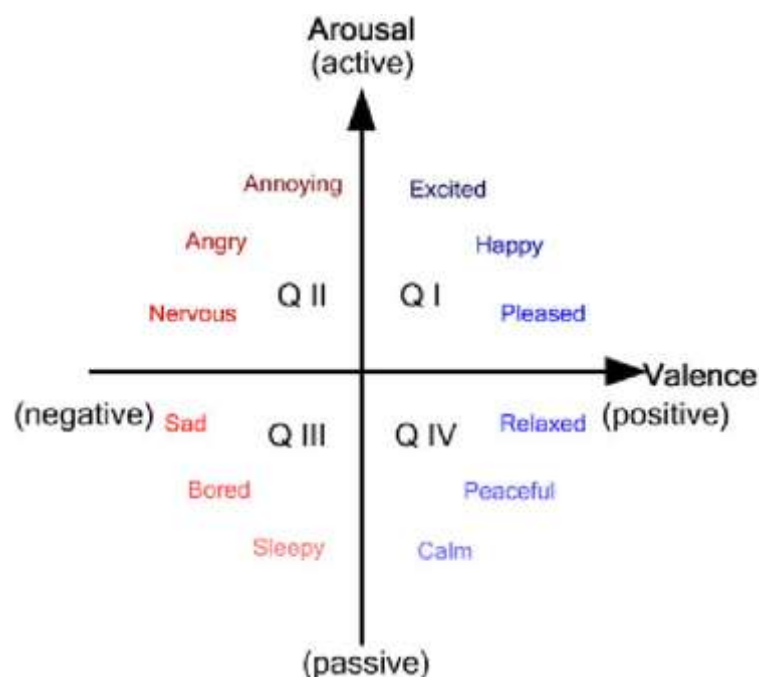
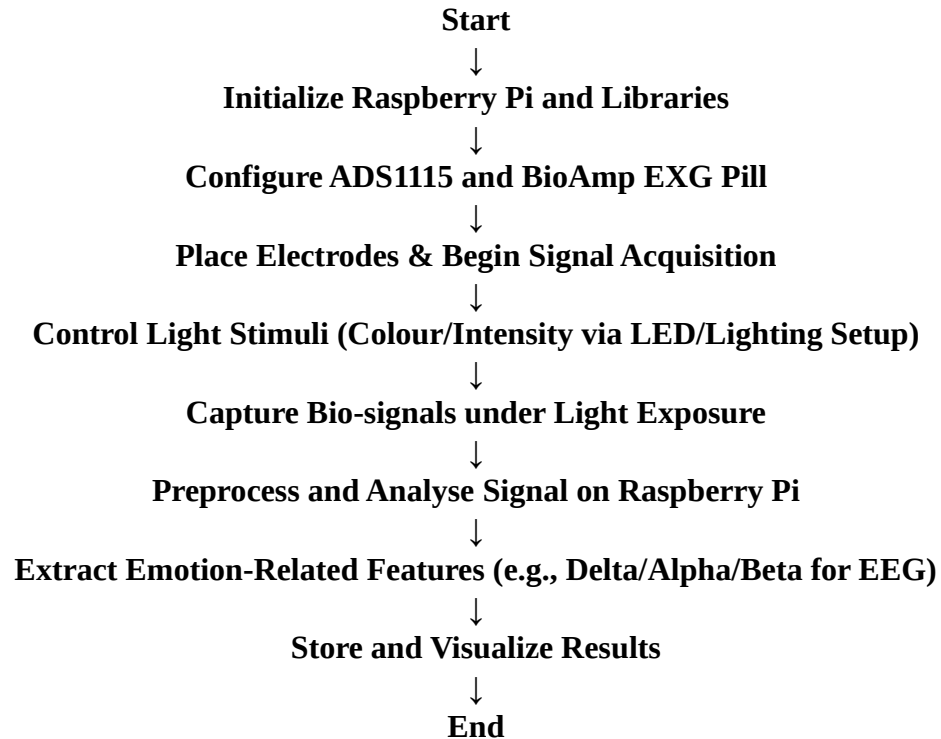


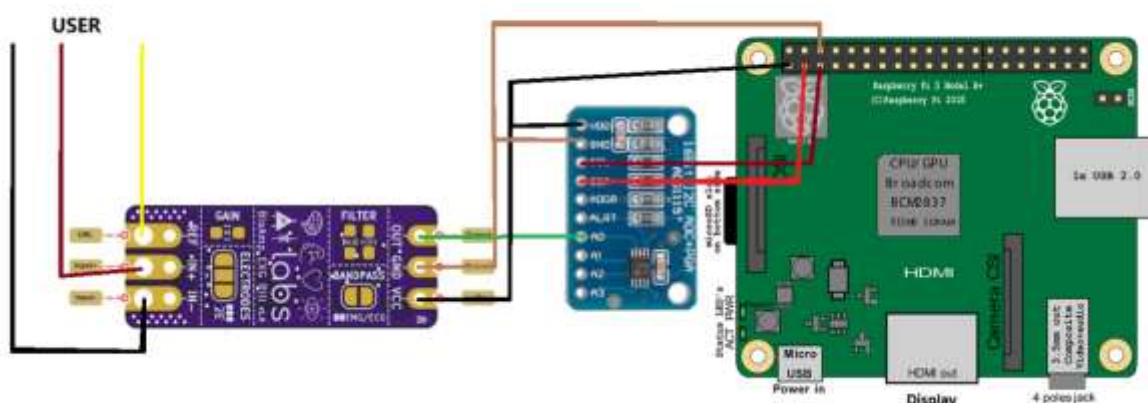
Fig 1: Arousal/Activation vs. Valence Emotional Spectrum

Our algorithm uses the digitized EEG signal to create an equivalent polar plot which is further divided into 12 equal sub-regions corresponding to each emotion. The algorithm calculates the percentage sample density on each section to formulate our results.

4. Flowchart:



5. Circuit Diagram:



BioAmp EXG Output	ADC A0
ADC SCL	Pin 5 - GPIO3 (SCL)
ADC SDA	Pin 3 - GPIO2 (SDA)

BioAmp EXG VCC	ADC VDD	Pin1/17 - +3.3V
BioAmp EXG GND	ADC GND	Pin 6 - GND

Fig 2: Circuit Diagram of our Prototype

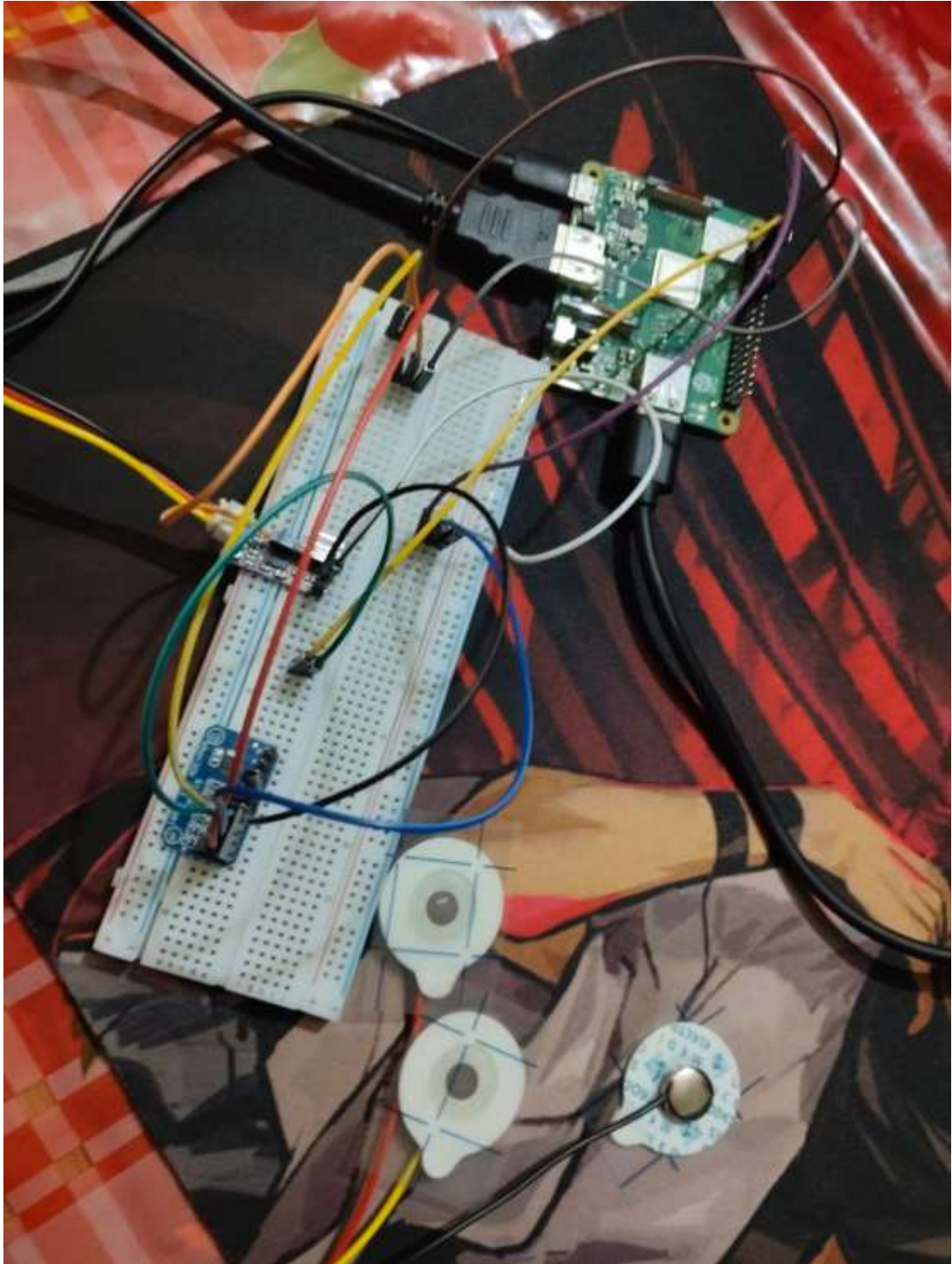


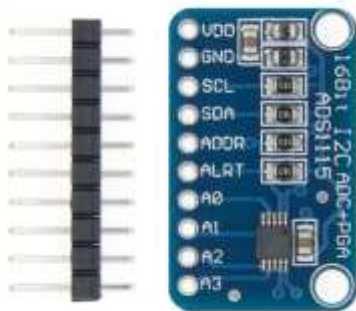
Fig 3: Prototype

6. Components:

COMPONENT	DESCRIPTION
BIOAMP EXG PILL	Analog front-end for bio-signal acquisition (EEG). Compact, open-source.
ADS1115	16-bit ADC with 4 channels and programmable gain; interfaces via I2C.
RASPBERRY PI 3A+	Small-form-factor Linux computer; handles signal processing, I2C control, and UI.
ELECTRODES	Non-invasive gel electrodes for signal acquisition from the skin surface.
LIGHT SOURCE	White/RGB LED or bulb to simulate different light environments.
WIRES & BREADBOARD	Standard prototyping hardware for connections.

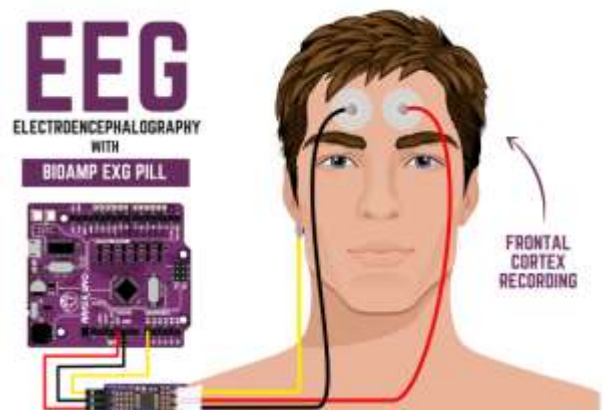


Raspberry Pi 3A+ is used to process the signal via Python and give us the output.



ADS1115 Analog to Digital Converter is used to convert the analog output of the EEG signal into a digital bitstream to be read by the Raspberry Pi.

EXG Pill reads the EEG of the user as an analog signal.



7. Results and Inference:

- Our prototype was run with two survey participants.
- Each participant was subjected to three lighting conditions.
- The following results were obtained:

1) Participant 1:

<i>Colour of Light/ Mood</i>	White (Control)	Red	Green
Pleased	7.4%	8%	7%
Happy	7%	7.6%	10%
Excited	8.2%	8.8%	6.8%
Annoyed	6.2%	7.4%	8%
Angry	9%	9.2%	7.4%
Nervous	6.6%	6.6%	10.2%
Sad	9.6%	9.2%	7.2%
Bored	8.6%	8.6%	9%
Sleepy (Tired)	10.4%	9.2%	10.2%
Calm	10.4%	8%	7.8%
Peaceful	9.6%	9.8%	8.6%
Relaxed	7%	7.6%	7.8%

2) Participant 2:

<i>Colour of Light/ Mood</i>	White (Control)	Red	Green
Pleased	8.5%	3.5%	4.75%
Happy	7.25%	7%	5.25%
Excited	6%	8.25%	9.5%
Annoyed	5.5%	6.75%	8%
Angry	9.75%	9%	7.25%
Nervous	10%	10.5%	6%
Sad	4%	5.5%	10%
Bored	11%	11.25%	11.5%
Sleepy (Tired)	8%	8%	8.25%
Calm	6.5%	6%	10.25%
Peaceful	12%	12%	9%
Relaxed	11.5%	12.25%	10.25%


```

srijon@raspberrypi:~$ python3 emotionpercentg.py
collecting 500 samples.....
EEG signal Emotion Mapping
pleased :37 samples(7.400000%)
happy :35 samples(7.000000%)
excited :41 samples(8.200000%)
annoyed :31 samples(6.200000%)
angry :45 samples(9.000000%)
nervous :33 samples(6.600000%)
sad :48 samples(9.600000%)
bored :43 samples(8.600000%)
sleepy :52 samples(10.400000%)
calm :52 samples(10.400000%)
peaceful :48 samples(9.600000%)
relaxed :35 samples(7.000000%)
srijon@raspberrypi:~$

```

Person 1: Test case white

```

srijon@raspberrypi:~$ python3 emotionpercentg.py
collecting 500 samples.....
EEG signal Emotion Mapping
pleased :40 samples(8.000000%)
happy :38 samples(7.600000%)
excited :44 samples(8.800000%)
annoyed :37 samples(7.400000%)
angry :46 samples(9.200000%)
nervous :33 samples(6.600000%)
sad :46 samples(9.200000%)
bored :43 samples(8.600000%)
sleepy :46 samples(9.200000%)
calm :46 samples(9.200000%)
peaceful :49 samples(9.800000%)
relaxed :38 samples(7.600000%)
srijon@raspberrypi:~$

```

Person 1: Test case red

```

srijon@raspberrypi:~$ python3 emotionpercentg.py
collecting 500 samples.....
EEG signal Emotion Mapping
pleased :35 samples(7.000000%)
happy :50 samples(10.000000%)
excited :34 samples(6.800000%)
annoyed :40 samples(8.000000%)
angry :37 samples(7.400000%)
nervous :51 samples(10.200000%)
sad :36 samples(7.200000%)
bored :45 samples(9.000000%)
sleepy :51 samples(10.200000%)
calm :39 samples(7.800000%)
peaceful :43 samples(8.600000%)
relaxed :39 samples(7.800000%)
srijon@raspberrypi:~$

```

Person 1: Test case green

Thus, we can infer that white light leads to more mellowed emotional balance, whereas red and green are leading to more intense emotional states.

8. Conclusion:

This project demonstrates a novel, low-cost approach to studying the impact of light on human emotion through real-time analysis of bioelectrical signals. By integrating the BioAmp EXG Pill, ADS1115 ADC, and Raspberry Pi 3A+, we successfully correlated light stimuli with physiological responses. The system offers a portable, non-invasive solution for monitoring emotional states, paving the way for applications in mental health, adaptive lighting, and human-centric design. Our findings highlight the potential of accessible embedded systems in bridging neuroscience and environmental interaction, encouraging further exploration into emotion-aware technologies.

9. Future Scope of Application:

- **Smart Lighting Systems:** Integration with home automation to adjust lighting based on real-time mood detection.
- **Mental Health Monitoring:** Continuous passive tracking of emotional wellbeing using light-stimulus responses.
- **Workplace Optimization:** Dynamic lighting adjustments to maximize productivity and reduce stress.
- **VR/AR Integration:** Enhance immersive experiences with mood-based environmental lighting.
- **Clinical Studies:** Further research on seasonal affective disorder (SAD) and therapeutic light exposure.

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